

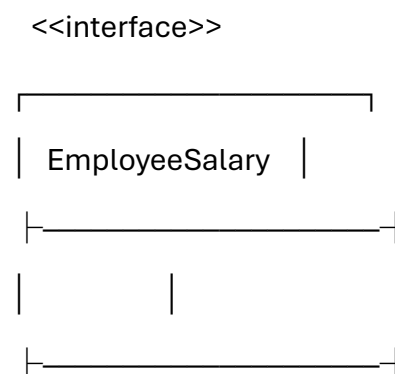
Aim

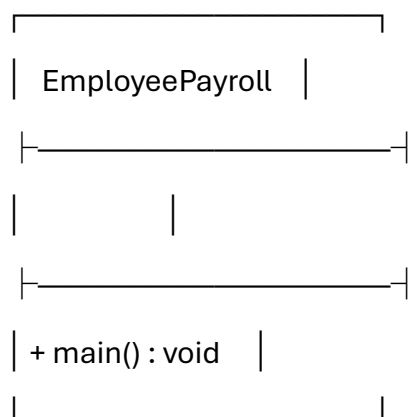
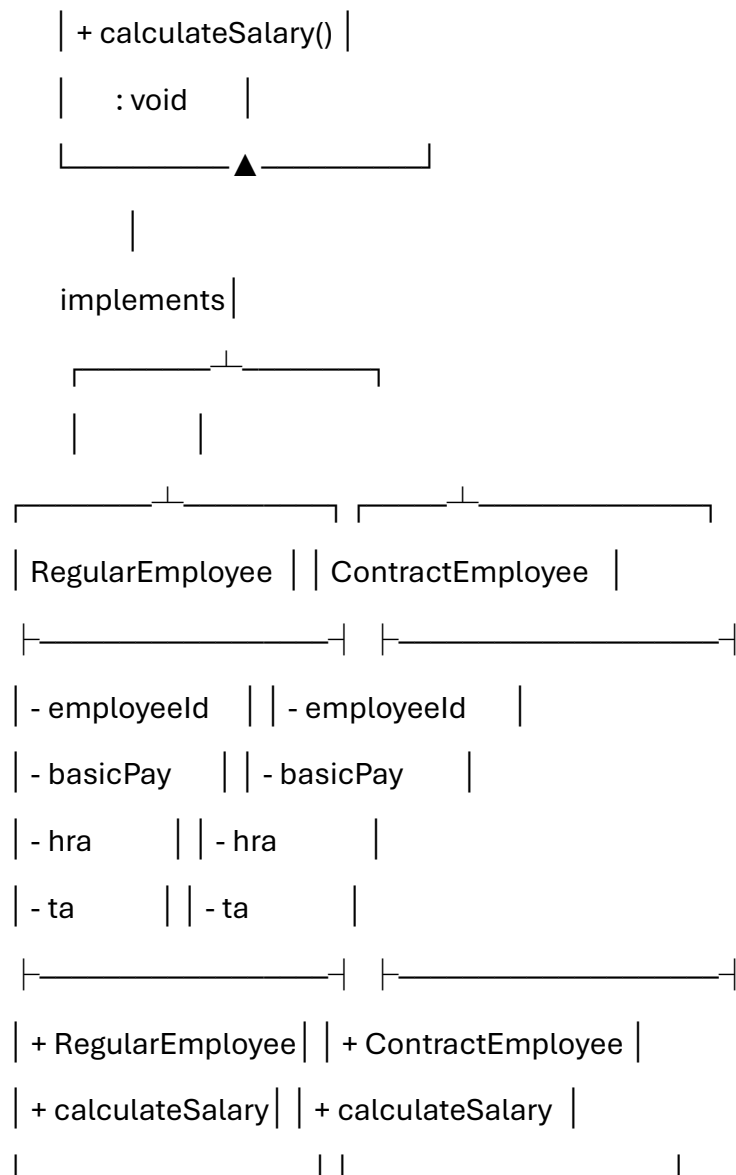
To write and execute a Java program using an **interface** to calculate and display the monthly salary details of **Regular Employees and Contract Employees**.

Algorithm

1. Start the program.
2. Define an interface named EmployeeSalary.
3. Declare the method calculateSalary() in the interface.
4. Create a RegularEmployee class that implements the interface.
5. Set Basic Pay = ₹25,000, HRA = ₹15,000 and TA = ₹5,000.
6. Calculate the total salary.
7. Create a ContractEmployee class that implements the interface.
8. Set Basic Pay = ₹12,000 and TA = ₹3,000, with no HRA.
9. Calculate the total salary.
10. Read the Employee ID from the user.
11. If the ID starts with R, create a RegularEmployee object.
12. Otherwise, create a ContractEmployee object.
13. Access the implementation using an **interface reference**.
14. Display the salary details.
15. Stop the program.

UML Diagram:





Code:

Package Records;

import java.util.Scanner;

// Interface

interface EmployeeSalary {

void calculateSalary();

}

// Regular Employee

class RegularEmployee implements EmployeeSalary {

private String employeeId;

private double basicPay = 25000;

private double hra = 15000;

private double ta = 5000;

RegularEmployee(String employeeId) {

this.employeeId = employeeId;

}

@Override

public void calculateSalary() {

double total = basicPay + hra + ta;

System.out.println("\nSalary Details:");

System.out.println("Employee Id: " + employeeId);

```
        System.out.println("Basic Pay: " + basicPay);  
        System.out.println("HRA: " + hra);  
        System.out.println("T.A: " + ta);  
        System.out.println("Total Amount: " + total);  
    }  
}
```

// Contract Employee

```
class ContractEmployee implements EmployeeSalary {
```

```
    private String employeeId;  
    private double basicPay = 12000;  
    private double hra = 0;  
    private double ta = 3000;
```

```
    ContractEmployee(String employeeId) {  
        this.employeeId = employeeId;  
    }
```

@Override

```
    public void calculateSalary() {
```

```
        double total = basicPay + hra + ta;
```

```
        System.out.println("\nSalary Details:");  
        System.out.println("Employee Id: " + employeeId);  
        System.out.println("Basic Pay: " + basicPay);  
        System.out.println("HRA: " + hra);
```

```
        System.out.println("T.A: " + ta);  
        System.out.println("Total Amount: " + total);  
    }  
}
```

// Main class

```
public class Interface {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Employee Id: ");  
        String id = sc.nextLine();  
  
        // Interface reference  
        EmployeeSalary employee;  
  
        if (id.startsWith("R")) {  
            employee = new RegularEmployee(id);  
        } else {  
            employee = new ContractEmployee(id);  
        }  
  
        employee.calculateSalary();  
  
        sc.close();  
    }  
}
```

Output:

```
ers\Sahithi py1a\AppData\Roaming\Code\User\workspaceStorage\234db3eb4e599b5ec014a58a5cc45c1
Enter Employee Id: 123

Salary Details:
Employee Id: 123
Basic Pay: 12000.0
HRA: 0.0
T.A: 3000.0
Total Amount: 15000.0
PS C:\Users\Sahithi py1a\Desktop\Sahithi java>
```